

Prenatal Exposure to Organophosphate Pesticides and IQ in 7-Year-Old Children

Context: Organophosphate (OP) pesticides are neurotoxic at high doses. Few studies have examined whether chronic exposure at lower levels could adversely affect children's cognitive development. **Objective:** We examined associations between prenatal and postnatal exposure to OP pesticides and cognitive abilities in school-age children. **Methods:** We conducted a birth cohort study (Center for the Health Assessment of Mothers and Children of Salinas study) among predominantly Latino farmworker families from an agricultural community in California. We assessed exposure to OP pesticides by measuring dialkyl phosphate (DAP) metabolites in urine collected during pregnancy and from children at 6 months and 1, 2, 3.5, and 5 years of age. We administered the Wechsler Intelligence Scale for Children, 4th edition, to 329 children 7 years of age. Analyses were adjusted for maternal education and intelligence, Home Observation for Measurement of the Environment score, and language of cognitive assessment. **Results:** Urinary DAP concentrations measured during the first and second half of pregnancy had similar relations to cognitive scores, so we used the average of concentrations measured during pregnancy in further analyses. Averaged maternal DAP concentrations were associated with poorer scores for Working Memory, Processing Speed, Verbal Comprehension, Perceptual Reasoning, and Full-Scale intelligence quotient (IQ). Children in the highest quintile of maternal DAP concentrations had an average deficit of 7.0 IQ points compared with those in the lowest quintile. However, children's urinary DAP concentrations were not consistently associated with cognitive scores. **Conclusions:** Prenatal but not postnatal urinary DAP concentrations were associated with poorer intellectual development in 7-year-old children. Maternal urinary DAP concentrations in the present study were higher but nonetheless within the range of levels measured in the general U.S. population.